FSM States

Challenge 2, NAKUL KADAM

All states used (Mealy + Moore) : in WXYZ, order of receive is W,X,Y,Z (Z being the most recently received bit)

State	Represents	Mealy	Moore
START	X (0 or 1)	4’d0	5’d0
S0	0	4’d1	5’d1
S1	1	4’d2	5’d2
S00	00	4’d3	5’d3
S01	01	4’d4	5’d4
S10	10	4’d5	5’d5
S11	11	4’d6	5’d6
S000	000	4’d7	5’d7
S001	001	4’d8	5’d8
S010	010	4’d9	5’d9
S011	011	4’d10	5’d10
S100	100	4’d11	5’d11
S101	101	4’d12	5’d12
S110	110	4’d13	5’d13
S111	111	4’d14	5’d14

State	Represents	Mealy	Moore
Sh0	0000	–	5’d15
Sh1	0001	–	5’d16
Sh2	0010	–	5’d17
Sh3	0011	–	5’d18
Sh4	0100	–	5’d19
Sh5	0101	–	5’d20
Sh6	0110	–	5’d21
Sh7	0111	–	5’d22
Sh8	1000	–	5’d23
Sh9	1001	–	5’d24
ShA	1010	–	5’d25
ShB	1011	–	5’d26
ShC	1100	–	5’d27
ShD	1101	–	5’d28
ShE	1110	–	5’d29
ShF	1111	–	5’d30

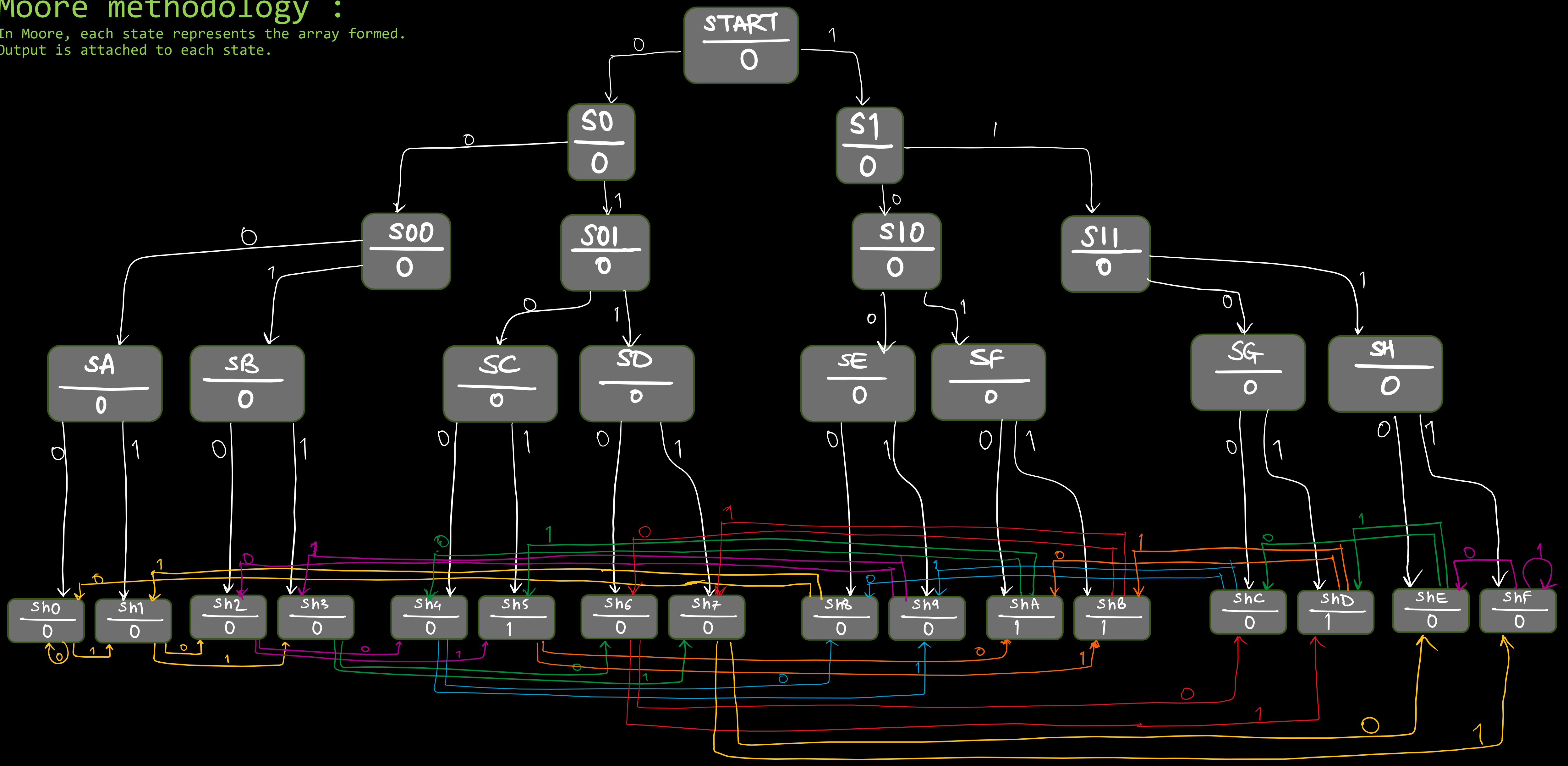
Moore states explained : (rightmost is the most recently received bit)

State	Represents	Explanation
START	X (0 or 1)	The initial/reset state Ready to receive input bits
Single digit States S0, S1 e.g S1	1	Only one digit received till now START → S1
Double digit States S00 to S11 e.g. S01	01	Two digits received till now State transitions : START → SO → S01
Triple digit States S000 to S111 e.g. S110	110	Three digits received till now State transitions : START → S1 → S11 → S110
Four digits States Sh0 to ShF e.g. Sh7	0111	Machine has received four bits by now. Now it will keep shuffling between these states till Reset is pushed State transition to reach : START → S0 → S01 → S011 → Sh7 After this state, if in=0 : Sh7 → ShE (ShE=1110) After this state, if in=1 : Sh7 → ShF (ShF=1111)

Moore State Transition Diagram :

Moore methodology :

In Moore, each state represents the array formed.
Output is attached to each state.



out=1 when: State either of Sh5, ShA, ShB, ShD.

Mealy states explained : (rightmost is the most recently received bit)

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Double digit States S00 to S11 e.g. S01	01	Two digits received till now State transitions : START ⟶ SO ⟶ S01
Triple digit States S000 to S111 e.g. S110	110	These states start after the 3rd bit from START is received. Once reached here, the 4bit sequence for every cycle is seen as XYZ|in; then the 3digits for next state becomes YZin. Possible State transitions : 1) START ⟶ S1 ⟶ S11 ⟶ S110 2) S011 & in=0 ⟶ S110

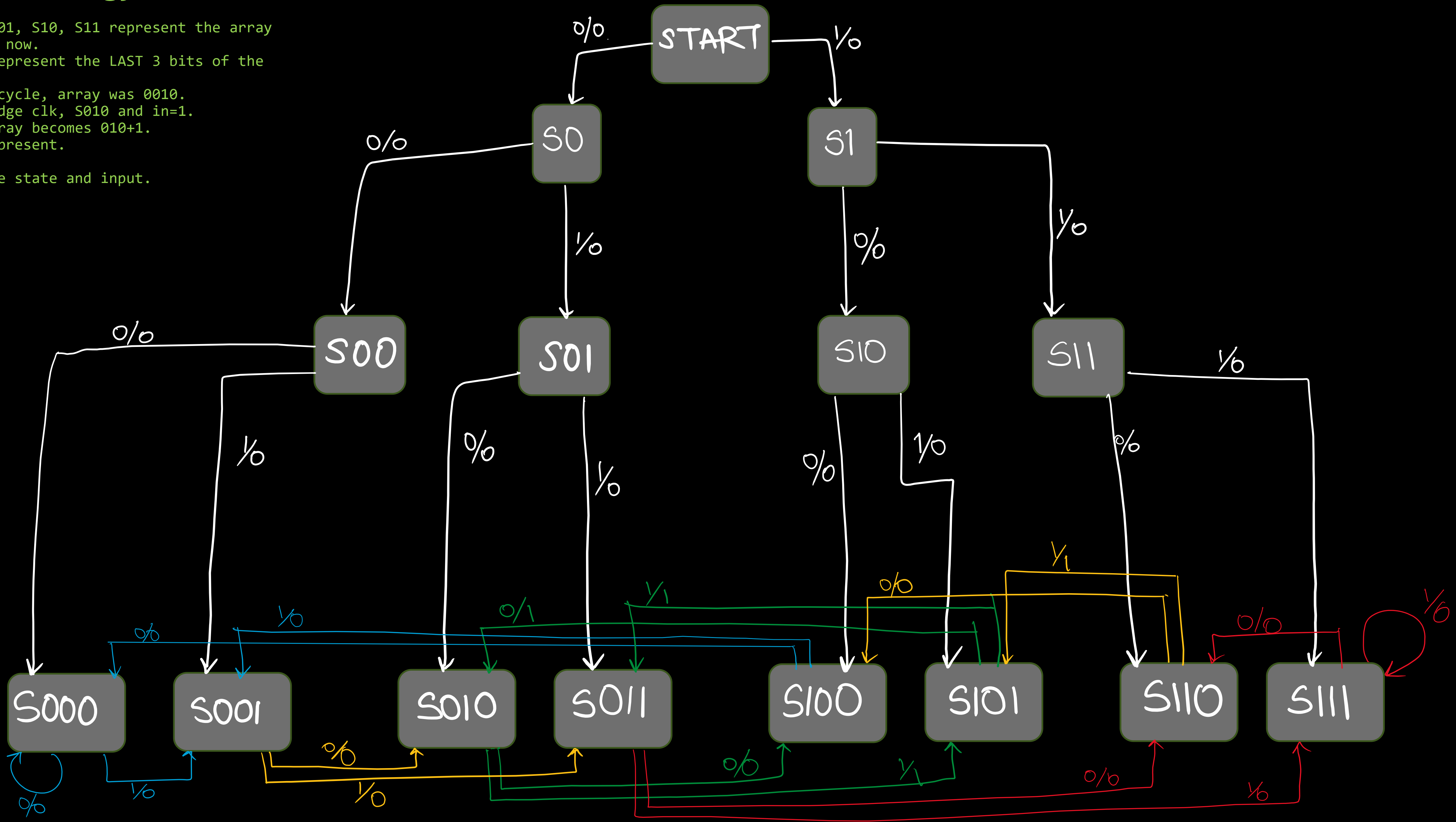
Mealy State Transition Diagram :

Mealy methodology :

In Mealy,
States S0, S1, S00, S01, S10, S11 represent the array of digits formed till now.
States S000 to S111 represent the LAST 3 bits of the current 4-bit array.
e.g. Say for one clk cycle, array was 0010.
     So, at next posedge clk, S010 and in=1.
     Current 4-bit array becomes 010+1.
     So out=1 as 101 present.

Output is based on the state and input.

★ NOTE : representation is input/output



FRAME PACKET RECEIVER states:

State	Represented as	Explanation
IDLE	2’d0	Waiting for the start bit 0 Jump to DATA if start bit is received
DATA	2’d1	Read the 8 data bits for 8 cycles Jump to PARITY on the 9th cycle
PARITY	2’d2	Read the Parity bit Jump to STOP
STOP	2’d3	Read the Stop bit If Stop bit isn’t 1, set frame_err If parity received in PARITY is not the same as calculated parity, set parity_err If both errors don’t occur, output 8 bits are the 8 data bits received. Else, output data bits reset to 8’d0 Set done to 1’b1 to indicate reception completed Jump back to IDLE

FRAME PACKET RECEIVER state transition diagram: